



# Tecnológico de Monterrey

**Campus Santa Fe**

**Dimension Deck GDD**

**Course:**

Software Construction and Decision Making

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# 1. Game Design

## 1.1 Summary

A retro-styled open-world adventure game inspired by the classic Atari aesthetic, where every room of the map hides a surprise: a new card to collect, a hidden reward, or an unexpected boss fight. The player builds their deck as they explore, finding attack, defense, and support cards scattered across the world, creating a unique and evolving strategy with every run. Combat is entirely card-based, meaning the hand you've built determines whether you survive what's behind the next door.

With a world that shifts themes and challenges as the player progresses and a randomized structure that ensures no two runs ever feel the same, the game delivers the tension of a roguelite wrapped in the charm of classic pixel-art adventure.

Dimension Deck sits at the intersection of two genres without belonging fully to either. Unlike *Slay the Spire*, where combat is turn-based and purely strategic, and unlike *Neon White*, where cards are disposable movement tools, Dimension Deck makes cards and real-time movement equally necessary at all times. You cannot win by standing still and thinking, and you cannot win by running without a plan. The closest comparison is: *The Binding of Isaac* meets *Slay the Spire*, but where your deck actively fights back through automated synergies without you pressing a button.

## 1.2 Gameplay

This is a dynamic game that combines a fast-paced action roguelite with the dynamic freedom of a deck builder using a 2D design. It uses the movement, top-down from *The Binding of Isaac*, with strategic deck management that *Slay the Spire* uses.

The players use their deck of cards in order to materialize attacks, defenses, and effects performed at the moment. The combat mechanics consist of two different types of cards. Active cards that the user can use in order to attack, heal or defend; and secondary cards that help the user when certain conditions are met or that help as stat boosts for the player.

We let the player navigate in randomized rooms in order to find loot, shops, or enemies, leading to the main goal of advancing from rooms and worlds by clearing rooms. We let the user draft new cards by buying them or obtaining them by killing enemies and bosses with a final purpose of killing the final boss of each room in order to advance.

### **Generation Rulebook**

Each dimension is built from a sequence of rooms generated at the start of every run. A dimension always contains a minimum of 8 rooms and a maximum of 12 rooms before a Boss node is placed at the end. The exact number is chosen randomly within that range each time, so the player never knows how close they are to the boss until they reach it.

Rooms are not placed in a straight line. The generator builds a small branching map where the player may have two or three possible paths to take at certain points, but all paths eventually converge at the Boss node. This means two players can go through the same dimension in the same run count and experience a different number and type of rooms depending on which branch they follow.

### **Room Type Weighting**

Not every room is a combat room. The generator pulls from the following pool when placing each room:



| Room Type   | Base Chance | Notes                                             |
|-------------|-------------|---------------------------------------------------|
| Combat Room | 60%         | Always available                                  |
| Chest Room  | 15%         | Drops one card on clear                           |
| Shop Room   | 10%         | Increases over time — see Shop Room Scaling below |
| Shrine Room | 10%         | Offers one upgrade trade                          |
| Glitch Room | 5%          | Contains a Glitch Pillar                          |

**Shop Room Scaling:** The chance of a Shop Room appearing increases by 10% for every 3 consecutive combat rooms the player clears without visiting a shop. This bonus stacks and resets to 0% the moment a Shop Room is entered. This prevents the player from going deep into a dimension without any access to cards or upgrades.

## 1.3 Mindset

We want the player to feel overwhelmed by the number of enemies on screen but empowered by the automated synergies their deck produces. The tension comes from managing movement and card rotation at the same time. A good run should feel like controlled chaos, where the player is always one smart card play away from turning a losing fight around. The emotional target is: *"I was about to die, but my deck saved me."*

## 2. Technical

### 2.1 Screens

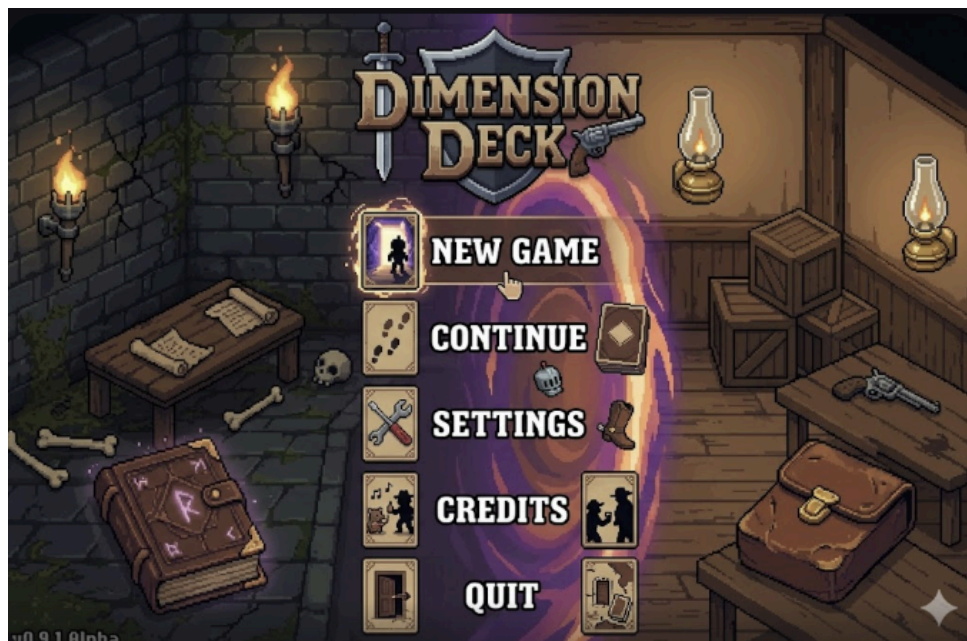
Dimension Deck uses different core screens. Each of them has their own identity and utility for the core gameplay, being important for the well-developed game. We divided the

screens into three types, being navigation and progression screens, gameplay and customization screens, and the economy and result screens.

## Navigation and progression screens

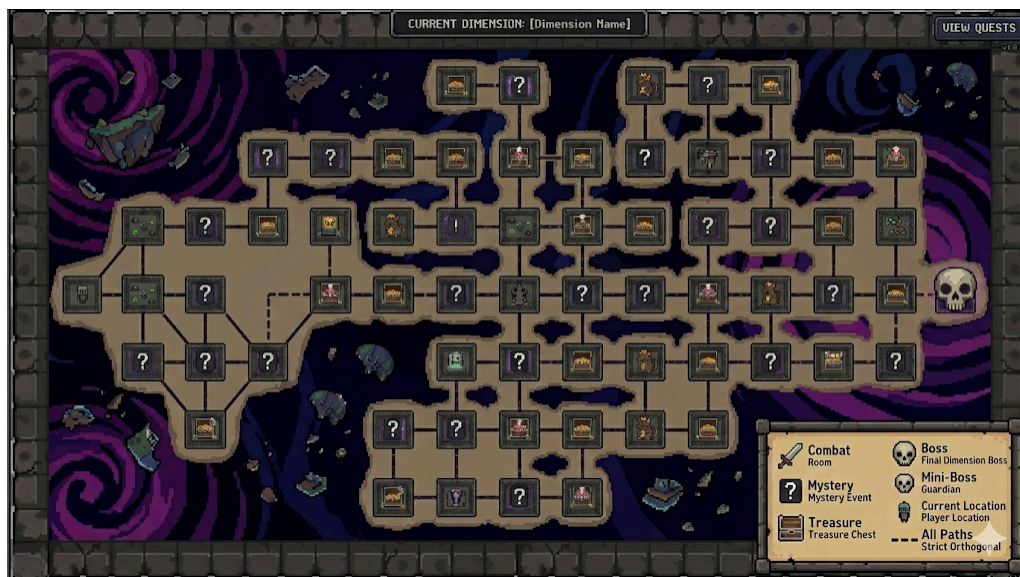
### Main Menu

- Purpose: The main screen of our game handles the login session of the player, the settings, credits and the ability for the user to quit the game.
- Elements: New Game, Continue, Settings/Options, Credits, and Quit.
- Visuals: Game logo, version number, and thematic background art.



### Dimension Map

- Purpose: A whole view of the dimension of the current run, allows the player to choose their own path to follow and level.
- Elements:
  - Nodes: Icons represent the type of the level (Combat, Event, Treasure, Boss).
  - Pathing: Lines showing valid movement options between nodes.
  - Legend: A guide explaining what each map symbol means.



## Gameplay and customization screens

### Room Screen

- Purpose: The screen in which the combat takes place
- Elements:
  - Player/Enemy Stats: Health bars, status effects, and resource meters.
  - Active Hand: The current hand deck of the player.



### Deck Management Screen

- Purpose: A screen for the player to manage their deck and view their collection.
- Elements:
  - Grid View: All cards currently in the player's deck.
  - Filter/Sort: Options to view by cost, rarity, or type.
  - Stats Sidebar: Shows deck archetypes, average cost, and total card count.
  - Action Buttons: Remove Card (if at a service), Upgrade Card, or Save Deck.



## Economy and result screens

### Shop Screen

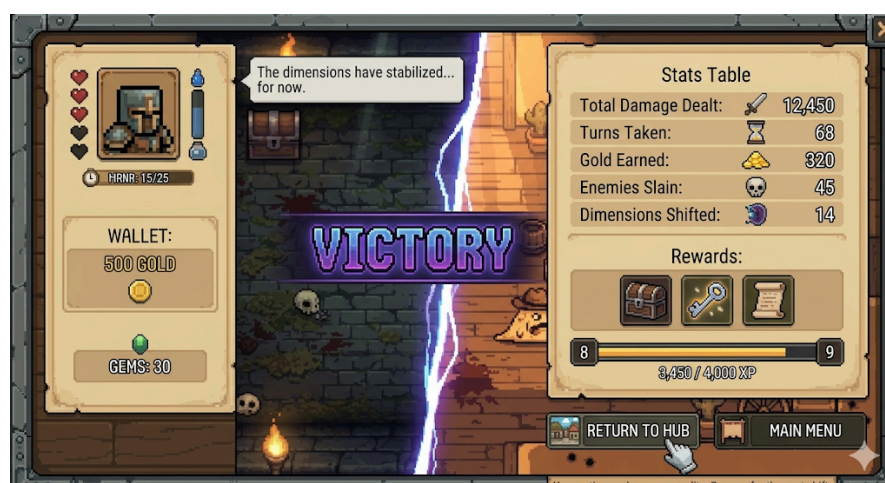
- Purpose: To exchange earned currency for active cards, secondary cards or permanent purchase.
- Elements:
  - Inventory Slots: Items for sale with price tags clearly displayed.
  - Wallet: A prominent display of the player's current gold/currency.
  - Flavor Text: Dialogue from the merchant NPC.
  - Buttons: Buy, Sell (if applicable), and Leave Shop.





### Game Over / Victory Screen

- Purpose: Summarize the stats the player made during that run, and display the outcome of it.
- Elements:
  - Summary Text: "Victory" or "Defeated" in large.
  - Stats Table: Total damage dealt, turns taken, gold earned, and enemies slain.
  - Rewards: Loot chests or XP bars filling up.
  - Buttons: Return to Hub (Victory) or Main Menu (Game Over)





## 2.2 Controls

### Input Mapping

#### Movement and Navigation

- W: Move the player character upward.
- A: Move the player character to the left.
- S: Move the player character downward.
- D: Move the player character to the right.
- Arrow Up: Move the player character upward (alternative to W).
- Arrow Down: Move the player character downward (alternative to S).
- Arrow Left: Move the player character to the left (alternative to A).
- Arrow Right: Move the player character to the right (alternative to D).
- Mouse Cursor: Aim and target enemies or card destinations during combat.
- M (tap): Toggle the mini-map overlay in the corner of the screen.
- M (hold): Open the full-screen tactical map showing the entire dimension layout, including room types and the path to the Boss node.

#### Card Actions

- Key 1: Execute the first card in the active hand (leftmost slot).
- Key 2: Execute the second card in the active hand.
- Key 3: Execute the third card in the active hand.
- Key 4: Execute the fourth card in the active hand (requires slot expansion).
- Key 5: Execute the fifth card in the active hand (requires slot expansion).
- Left Click (LMB): Confirm and execute the currently selected card at the cursor's location.

### **Combat and Utility**

- Spacebar: Dodge or dash in a quick burst in the direction the player is currently moving.
- E: Interact with objects such as chests, Glitch Pillars, Dimensional Altars, the Card Smith's Anvil, and the shop.
- Tab: Toggle the deck management overlay to review and organize cards during exploration.
- Escape: Pause the game or close any open menu.

### **Technical UX Integration**

- Smart Mapping: When a card is played, the numbers shift left to fill the gap, ensuring no "dead keys" in the middle of a hand.
- Map Persistence: Flag rooms are permanently revealed once the player enters the collider for that room, even if they backtrack to a previous floor.
- UI Tooltips: In the deck management screen (tab), hovering over a card with the mouse displays its full stats and lore, keeping the combat HUD clean.

## 2.3 Cards

### 2.3.1 Card System

Cards are the key for combat and progression in Dimension Deck. Every action in the game is executed through the players deck. The deck of the player is divided into two different categories.

#### Active Cards (Manual Execution)

Active cards are tools that the player triggers manually. These cards are the weapon of the player in order to deal damage or generate healing or defense depending on the player's decision.

- Capacity: You begin with 3 slots, expandable to 5.
- The Rotation: Once played, the card goes into a cooldown period before returning to your hand. This prevents the player to spam the same card and learn when to use them wisely.

#### Automatic Cards (Trigger-Based)

Automatic cards are the secondary hand of the player. This one activates depending on certain conditions; if these ones are met, the card will be activated.

- Capacity: Supports between 4 and 8 slots.
- Function: These cards reward synergistic deck-building.

#### Card Rarities

| Rarity | Color  | Description                                    |
|--------|--------|------------------------------------------------|
| Common | Bronze | Base-level cards that can be found frequently. |



|                  |        |                                                                                      |
|------------------|--------|--------------------------------------------------------------------------------------|
| <b>Rare</b>      | Blue   | Better cards with combination effects or specialized utility.                        |
| <b>Epic</b>      | Purple | Powerful mechanics that often serve as the cornerstone of a specific build strategy. |
| <b>Legendary</b> | Gold   | Game-changing finds that fundamentally alter how you approach every run.             |

### The Card Resistance System

Expanding the players' deck does two things. Increases the power of the player, but also a card resistance system activates. As more slots of cards are upgraded by the player, enemies gain resistances, forcing the player to change their tactics and keep evolving through the run.

### Card Resistance Scaling Formula

Resistance % = (Active Card Slots – 3) × 5%

At the starting 3 slots: 0%. At 4 slots: 5%. At 5 slots (max): 10%.

The automatic pool carries its own separate resistance on top of that. For every 2 automatic slots above the base 4, enemies gain an additional 3%.

| Active Card Slots | Card Resistance Applied |
|-------------------|-------------------------|
| 3 (base)          | 0%                      |
| 4                 | 5%                      |
| 5 (max)           | 10%                     |

The same formula applies to the automatic card pool but uses a separate multiplier. For every 2 automatic slots above the starting 4, enemies gain an additional 3% resistance on top of the base value from active slots.

| Automatic Card Slots | Additional Resistance |
|----------------------|-----------------------|
| 4 (base)             | +0%                   |
| 6                    | +3%                   |
| 8 (max)              | +6%                   |

Both resistance values stack. A player with 5 active slots and 8 automatic slots faces a total of  $10\% + 6\% = 16\%$  damage reduction on all enemies in the current room.

### 2.3.2 Synergy System

The synergy system rewards sequencing over raw damage. A trigger card sets a flag on an enemy. If a matching catalyst card hits that same enemy within 4 seconds, the combination fires with a damage multiplier. Miss the window the flag clears and you get nothing.

#### Current defined combinations:

| Trigger Card    | Catalyst Card   | Result    | Radius        | Duration  | Dmg Multiplier |
|-----------------|-----------------|-----------|---------------|-----------|----------------|
| Oil (Automatic) | Fire (Active)   | Area Burn | 3 tiles       | 2 seconds | ×3             |
| Freeze (Active) | Strike (Active) | Shatter   | Single target | Instant   | ×2.5           |

|                       |               |             |         |           |    |
|-----------------------|---------------|-------------|---------|-----------|----|
| Poison<br>(Automatic) | Wind (Active) | Toxic Cloud | 2 tiles | 3 seconds | ×2 |
|-----------------------|---------------|-------------|---------|-----------|----|

The Synergy Manager holds the trigger flag for exactly 4 seconds. A larger deck means more possible combinations, but also more moving parts to sequence correctly while dodging in real time. The skill is not memorizing the table; it's knowing when you actually have a window to land both cards.

### 2.3.3 Card Acquisition

#### Field Rewards (Exploration)

Cards found in the environment reward thoroughness and combat prowess:

- Combat: Clearing the final boss of each level grants the player a choice of 3 cards from a randomized selection to help the player keep improving their deck.
- Treasure Chests: Hidden throughout procedural floors, chests contain randomized cards. Players who explore every corner of each level are more likely to find synergy cards.

#### The Dimension Store (Shop)

The shop serves as the primary hub for managing the deck of the player:

- Credits: Money use in the game in order to purchase cards in the store. This credits are dropped by enemies or chests.
- Store: Designed places that offer a rotating stock of cards, allowing players to find specific cards to complete a deck archetype.
- Removal Services: Beyond buying cards, the store provides the critical ability to purge basic or unwanted cards from the deck, increasing the statistical consistency of high-tier plays.

## Glitch Pillars

These are hidden objects scattered across rooms that do not appear on every run.

- When the player touches a Glitch Pillar, it stabilizes and rewards them with a single card pulled randomly from any rarity tier.
- Because the card is random and the pillar only gives one option, the player cannot choose or refuse it; the card goes directly into the deck.
- Glitch Pillars are the only source of cards that bypass the three-option selection screen, making them unpredictable but potentially very rewarding.

## The Loop

This system ensures that no two runs are identical:

- Route Optimization: The player's path on the map indicates the potential card pool. A path heavy on combat nodes favors rapid acquisition, while a shop path allows for precision deck-building.
- Adaptive Strategy: Players must constantly change their strategy based on the cards they encounter during exploration versus those they can afford to purchase.

### 2.3.4 Card Upgrades

Every card has three levels: base, upgraded and max. When a card is obtained, it will start at a base level. The moment the player acquires a second copy of the same card, these two cards will merge automatically, advancing the level of the card into upgraded. Acquiring a third copy of the same card will trigger the same automatic merge, changing the card level from upgraded to max.

When a card reaches a max level and the player obtains another copy of the same card, this extra card will not be merge and instead it will transform into dimensional credits. The

amount of credits depends on the rarity of the card: common duplicates return 10 credits, rare return 25, epic return 50, and legendary return 100.

Each level of upgrade improves the card in a specific way rather than just increasing a number. The changes are defined per card type.

#### Active Cards

- Upgrade: Reduces the card's cooldown by 30%, letting the player use it more frequently before it enters the discard pile.
- Max: Increases the card's base damage, healing value or defense value by 40% and unlocks a secondary effect that did not exist at lower levels. For example, a Max-level Strike card not only deals more damage but also briefly slows the target after the hit.

#### Secondary Cards

- Upgrade: Widens the trigger condition, making the card activate in more situations than it did at base level.
- Max: Increases the potency of the effect itself, whether that means a larger damage bonus, a longer buff duration, or a stronger status applied to enemies.

### 2.3.5 Combat: Card Rules

Combat in Dimension Deck relies on constant attention to the scenario. Efficiency is measured by the ability of the player to cycle through their hand while maintaining good positioning for a real-life battle between enemies.

#### Active Card Execution

- Instant Activation: Pressing the corresponding number key or clicking the mouse triggers the card's effect immediately at the cursor's location. There is no "cast time" or mana cost, allowing for seamless integration with movement.

- The Cooldown State: Once a card is activated, that specific slot enters a cooldown state. The card face dims, and a circular "wipe" animation indicates the remaining recharge time. The player cannot use that slot again until the timer finishes.
- Variable Recovers: Cooldown durations are tied to card power and rarity.
  - Low-Tier/Common: Short cooldowns (e.g., 1–3 seconds), allowing for consistent, "spam-able" basic attacks or minor shields.
  - High-Tier/Legendary: Long cooldowns (e.g., 10–20 seconds), requiring the player to save these "Ultimate" abilities for boss phases or being overwhelmed.

### **Damage and Interaction Collision-Based Damage**

All card-based damage is physical and projectile-based. Damage is dealt only if the card's effect physically intersects with an enemy's hitbox.

- Directional Shielding: Defensive cards like the Wood Shield provide protection only in the direction the player is facing or aiming. Players must "active-block" by timing the card activation with the enemy's telegraphed strike.
- Scaling Impact: A card's damage and effect duration scale based on its Level and Rarity. Higher-level cards may introduce "Burst Damage" or "Stun Frames," allowing players to interrupt enemy wind-up animations.

### **Hand Cycling and Rotation Strategy**

Since there is no mana limit, the primary skill gap is rotation management. Players must cycle through their 1–5 keys to ensure they always have an offensive or defensive option available, avoiding "all-on-cooldown" moments where they are left defenseless.

#### **2.3.6 Type of Cards**

| # | Name         | Rarity | Cost | Effect                      |
|---|--------------|--------|------|-----------------------------|
| 1 | Quick Strike | Common | 1    | Deal 20 damage to one enemy |

|    |                  |           |      |                                                                                      |
|----|------------------|-----------|------|--------------------------------------------------------------------------------------|
| 2  | Area Blast       | Common    | 2    | Deal 15 damage in a 3-tile radius                                                    |
| 3  | Heal Pulse       | Common    | 2    | Restore 25 HP                                                                        |
| 4  | Oil Slick        | Common    | Auto | Apply Flammable to nearest enemy when HP drops below 50%                             |
| 5  | Iron Skin        | Rare      | Auto | Reduce next damage taken by 10, resets each room                                     |
| 6  | Freeze Shot      | Rare      | 2    | Freeze one enemy for 2 seconds                                                       |
| 7  | Toxic Spore      | Rare      | Auto | Apply Poison when player dodges through an enemy                                     |
| 8  | Chain Lightning  | Epic      | 3    | Hit one enemy for 25 damage; jumps to 2 nearby enemies for 12 each                   |
| 9  | Void Anchor      | Epic      | Auto | When entering a boss room, reduce boss Card Resistance by 20% for 10 seconds         |
| 10 | Dimensional Edge | Legendary | 4    | Deal 50 damage; if used in a different dimension than where acquired, damage doubles |

## 2.3.7 Card Design



Defense

Damage

Heal





## 2.4 Combat

### 2.4.1 Combat Overview: Real-Time Tactical Movement

Combat in Dimension Deck is defined as a real-time, continuous experience where enemies spawn and attack immediately upon the player entering a room. The system requires simultaneous attention to spatial awareness and card-based decision-making.

- **Movement and Defense:** Players use WASD or arrow keys for top-down directional control to physically dodge projectiles and melee strikes.
- **The Dodge Mechanic:** The spacebar triggers a dash, providing a reliable escape tool when the player is surrounded.
- **Card Execution:** Players aim with the mouse and execute active cards using number keys 1 through 5.
- **Telegraphed Danger:** Enemy attacks are telegraphed via wind-up animations or visible paths projected on the floor, giving the player seconds to react.

### 2.4.2 Enemy Behavioral Archetypes

All enemies are built upon three foundational archetypes that dictate their AI and physical presence:

- **Swarm Archetype**
  - **Characteristics:** These are small enemies, typically rendered at 16 x 16 px.
  - **Behavior:** They use a "seek-and-close" movement pattern to pursue the player.
  - **Combat Role:** They pressure the player by crowding the playable space, forcing constant movement to avoid contact damage.
- **Tank Archetype**
  - **Characteristics:** Large, high-health sprites that communicate a high level of threat.

- Behavior: They move slowly but are highly resistant to knockback.
- Combat Role: They act as obstacles that deal massive damage on contact, requiring the player to maintain distance.
- Ranged Archetype
  - Characteristics: Mid-sized enemies that prioritize staying away from the player.
  - Behavior: They maintain a safe distance and track the player to fire projectiles on a cooldown.
  - Combat Role: They force the player to dodge across the room while trying to find an opening to counter-attack.

### 2.4.3 Dimension-Specific Bestiary

#### The Dark Ages

| Enemy       | Archetype | Combat Stats & Behavior                                     |
|-------------|-----------|-------------------------------------------------------------|
| Dungeon Rat | Swarm     | Fast-moving; relies on numbers to corner the player.        |
| Slime       | Swarm     | Slows player movement or leaves hazards on the stone floor. |
| Skeleton    | Tank      | High HP; uses slow but heavy, telegraphed melee swings.     |

#### Old Western

| Enemy      | Archetype | Combat Stats & Behavior                                 |
|------------|-----------|---------------------------------------------------------|
| Desert Rat | Swarm     | Aggressive small pests found near dry bushes and cacti. |

|             |        |                                                                                    |
|-------------|--------|------------------------------------------------------------------------------------|
| Bandit      | Ranged | Strafes across the sand while firing pistol shots.                                 |
| Cactus Thug | Tank   | High defense; punishes players who get too close with needle-based contact damage. |

#### 2.4.4 Strategic Progression: Card Resistance

To ensure the game remains challenging, the Card Resistance system scales with the player's power. As the player expands their card slots or builds a larger deck, enemies gain resistance to specific card types, preventing a single strategy from dominating the entire run.

### 2.5 Roguelite Progression and Telemetry

Runs are independent, and permanent death resets the player back to the Hub World with their current deck lost, requiring a fresh start for the next attempt. However, dimensional credits earned during a run are carried over and can be spent on permanent upgrades that expand stats and maximum card capacity across future runs. These Hub upgrades ensure long-term empowerment without trivializing individual runs.

The game collects telemetry data across four areas to support ongoing balance work. Player performance tracking monitors run duration and the highest biome reached to identify difficulty spikes. Win/loss analytics record the frequency of Game Over versus Victory screens to gauge overall accessibility. Card and trigger usage analyzes which active cards are chosen most often and which automatic triggers prove most effective in combat. Economic balancing logs dimensional credit earnings to see whether players prioritize new cards, shop items, or permanent Hub upgrades. All collected data feeds back into dynamic scaling

adjustments to Card Resistance, ensuring enemies stay competitive as the player's deck capacity grows.

Three specific KPIs the game will track:

1. **Time per dimension:** Average minutes spent in Old West vs. Dark Ages per run. This identifies which biome causes players to stall or rush, pointing to pacing issues.
2. **Most discarded card:** Which active card gets removed from the deck most often at the shop. A card discarded frequently signals it underperforms relative to its cost.
3. **Run length at death:** The room number where most Game Over screens occur. This pinpoints difficulty spikes in the procedural layout.

### 3. Level Design

#### 3.1 Themes

##### 3.1.1 Ambience

Dimension Deck uses a 16-bit visual style. This design reminds players of classic gaming eras while providing a fluid and nostalgic experience. The game features two distinct dimensions, each with its own look.

- **The Old West:** This theme uses dusty deserts and frontier towns to create a sense of lawlessness and freedom. The wide-open space of the desert provides a feeling of freedom that stands in total contrast to the next world.
- **The Dark Ages:** This world is inspired by fantasy games like Dungeons and Dragons and takes place inside a castle and caverns. This setting makes the player feel trapped in a heavy, enclosed environment.

Using these two themes helps prevent the player from getting bored with the visuals. It also allows for unique creative additions in each biome without breaking the player's feeling of being in the game or the core gameplay experience.

### 3.1.2 Objects

#### 3.1.2.1 Ambient

**Player Character:** The player uses a **32 x 32 px** sprite. This larger size makes the hero stand out clearly against the background and enemies.

**Theme Elements:** Decorative items like desert plants for the West or stone pillars for the Dark Ages help ground the player in each dimension.

#### 3.1.2.2 Interactive

Each interactive object follows defined rules for how the player engages with it:

- **Card Reward Chests:** Appear after a room is fully cleared. Require no key. The player presses E to open and receives a choice of 3 cards drawn from the current biome's rarity pool. If the player skips the chest, it disappears when they exit the room.
- **The Dimension Store:** A safe zone; no enemies spawn here. The shopkeeper offers 4 cards for sale per visit, randomly pulled from common and uncommon pools. Prices range from 20 to 80 dimensional credits depending on rarity.
- **Glitch Pillars:** The player touches them to interact; no button press needed. Each pillar either reveals a hidden exit or upgrades one common card in the player's current hand to uncommon. Only one pillar upgrade is allowed per run.
- **The Card Smith's Anvil:** Requires two copies of the same active card. After merging, the card's cooldown is cut by 40%. The original two copies are consumed.

- **Dimensional Altars:** Removes one automatic slot permanently for that run. In exchange, all active card damage increases by 30% for the remainder of the run. This choice cannot be reversed.
- **Dimension Shift effect on objects:** When the player crosses into a new dimension, all chests and pillars from the previous dimension are removed. Credits and deck contents carry over; room objects do not.

### 3.1.3 Challenges

Dimension Deck's challenges aim to evaluate players' reflexes and their capacity to devise a card strategy amidst relentless pressure.

- **Real-Time Combat:** During each run the enemies move and attack the moment they see the player. With this mechanic, we make the player stay in constant alert to dodge or protect incoming projectiles and melee hits while also trying to play their cards in order to deal damage to the enemies.
- **Attack visualization:** Most of the enemies will have animations or visible paths on the floor, giving the player seconds for them to react and dodge the incoming attacks.
- **Card Resistance:** As the deck gets bigger and the player gets stronger, enemies will start to develop resistances to certain types of cards that the player has during that run. This mechanic stops the player from using the same strategy the whole game, forcing him to change the idea the player is focusing on for completing each level.
- **Boss Battles:** At the end of each level, the player must fight a boss that changes its behavior and attacks the player as the fight goes on. These fights test how well the player has built their deck and how good their skills at the game are.

- **Procedural Rooms:** Each room will be randomized, making the player never know if the next room will be full of dangerous enemies, traps, loot or the level boss to finish the level and continue the run.
- **Permanent Death:** If you run out of health, the run ends and you lose your current deck, sending you back to the start to try again with a fresh strategy.

### 3.2 Game Flow

The flow of Dimension Deck focuses on moving through a series of unpredictable rooms, where the main goal of the player is to survive and find an exit to the next challenge. Here is how the game progression works:

- **Start of the run:** A run begins in the Dark Ages or the Old West, making each run feel unique from any other. The player enters the first room with a starting deck, full directional control and free decision-making.
- **Clearing the rooms:** In order to move forward, players must clear the current level they are on, which can be done in different ways. This could be achieved either by eliminating the room's boss or finding a rift that acts as a gateway to the next area. Once the level is cleared, the room could drop rewards like the dimensional credits.
- **The final boss encounter:** The final boss of a dimension does not appear at a fixed level; it can be encountered randomly, making the player never know when they will face the most difficult enemy of the run. The difficulty of this encounter scales dynamically using the Card Resistance system, meaning the boss becomes tougher if the player has built a larger, more powerful deck.
- **Post-Level Store:** After successfully finishing a level, the player enters the Dimension Store. This is a safe area between levels where players can spend credits on new cards, buy more slots for their deck, and manage their primary and secondary card hands.

- **Deck Management and Storage:** Players are able to see and manage their deck at any time, but the store is the best place to reorganize it. If a player has more cards than their current slot allows, they can store them in their collection, though carrying a larger deck increases the resistance enemies have against their attacks.
- **Dimension Transition:** Once the final boss is defeated, a portal opens to transport the player into the next dimension. This transition moves the player into a different environment with new visuals, enemies, and cards.
- **The Death/Victory Cycle:** If the player loses all health, the run ends and the current deck is lost. However, any credits earned are carried back to the Hub World to purchase permanent stat and capacity upgrades, making the character stronger for the next attempt.

## 4. Development

The codebase is organized around a set of abstract base classes that define shared behavior without implementing it, keeping individual systems extensible and the overall architecture clean.

- **Entity:** The root class for anything that exists in the game world. Holds position, velocity, health, and a basic update/render cycle. All moving, interactable objects extend this.
- **Card:** Defines the structure every card must follow: a name, a rarity, an energy cost, an effect function, and a card type flag (active or automatic). The effect function takes a reference to the current combat state and resolves the card's outcome. Cards also carry a cooldown value used by the discard pile timer system.



- **Enemy:** Extends Entity and adds abstract methods for movement behavior, attack patterns, a Card Resistance value that scales with the player's deck capacity, and a drop table reference. Every enemy type implements these differently.
- **Room:** Defines the container structure for a playable space: fixed dimensions of 480 x 352 pixels across a 15 x 11 tile grid, spawn points, exit conditions, and lists of enemies and objects. Room subtypes inherit from this and override spawn logic and exit triggers.
- **Biome:** Groups a set of room types, an enemy pool, a visual theme identifier, and a boss reference into a single traversable dimension. Controls which rooms can appear on the Dimension Map for that biome and manages the Card Resistance multiplier applied to all enemies within it.
- **Relic:** Defines passive items with a trigger condition and an effect callback. The base class handles registration with the combat event bus; derived classes define what triggers them and what they do.
- **Synergy Effect:** Defines the structure for combination effects between cards. Holds a trigger card reference, a catalyst card reference, a multiplier value, and a result effect function. Registered with the Synergy Manager at runtime.

## 4.1 Derived Classes

Derived classes implement the concrete behavior defined by each abstract parent. Below are the primary derived types used across the game's systems.

- **Player (extends Entity):** The player character is rendered at 32 x 32 pixels and changes visual design depending on the active dimension. Adds the active card slot manager (3 slots, expandable to 5), the automatic card pool (4 to 8 slots), hand management, dimensional credit tracking, and the dodge mechanic. Handles collision with pickups

and interaction prompts for objects. Listens to the input manager and translates inputs into movement vectors and card plays.

- StrikeCard, AreaCard, HealCard (extend Card, active type): The three foundational active card types. StrikeCard resolves direct damage against a single target. AreaCard applies damage across a radius centered on the cursor position. HealCard restores a portion of the player's missing health and enters a longer cooldown than offensive cards to prevent it from replacing resource management.
- ConditionCard (extends Card, active type): An active card that applies a status effect to one or more enemies and queues a synergy flag in the SynergyManager. The Oil card is an example: it applies a flammable condition that the SynergyManager monitors, ready to trigger a tripled damage multiplier when a Fire active card resolves on the same target.
- TriggerCard (extends Card, automatic type): Sits in the automatic pool and watches the combat event bus for its trigger condition. When met, it fires its effect function without player input. Examples include granting a combat buff when entering a boss room, or creating a defensive explosion when the player takes critical damage.
- BoostCard (extends Card, automatic type): A permanent stat modifier for the duration of the run. Adds its value to a stat register when added to the deck and subtracts it on removal. Has no event trigger and no cooldown it contributes passively at all times.
- SwarmEnemy, TankEnemy, RangedEnemy (extend Enemy): The three base enemy behavioral archetypes. Enemy sprite dimensions vary by type: small swarm enemies render at 16 x 16 pixels and use a seek-and-close movement pattern; tank enemies are larger, move slowly, resist knockback, and hit for high damage on contact; ranged enemies maintain distance, track the player, and fire projectiles on a cooldown. All three archetypes change visual design depending on the active dimension.

- **BossEnemy** (extends **Enemy**): Adds a phase list, a phase transition trigger, and a secondary attack pattern that activates in later phases. Also holds a unique mechanic flag that activates a special rule for the boss fight. Boss sprites scale up significantly from standard enemy sizes to communicate their threat on screen.
- **CombatRoom**, **ShopRoom**, **ChestRoom**, **ShrineRoom** (extend **Room**): Each overrides the spawn logic and exit condition appropriate to its type within the fixed 480 x 352 pixel, 15 x 11 tile grid. **CombatRoom** spawns enemies from the biome pool and unlocks exits on full clear. **ShopRoom** spawns the merchant object and allows free exit after browsing. **ChestRoom** spawns one chest tied to a rarity roll and exits after interaction. **ShrineRoom** spawns one shrine and exits after the player accepts or declines the offer.
- **SynergyManager**: Singleton class that registers all **SynergyEffect** instances at startup and listens to card resolution events during combat. When a trigger card resolves and sets a synergy flag on a target, the **SynergyManager** holds the flag for a defined window. If a matching catalyst card resolves on the same target within that window, the multiplier is applied and the flag is cleared.
- **TelemetryLoggerPassive** singleton that hooks into key game events without affecting gameplay logic. Logs run duration, biome reached on death, Game Over and Victory frequencies, card selection rates, automatic trigger activation rates, and dimensional credit spending patterns. Data is written to a local session file and can be exported for balance review.

## 5. Graphics

### 5.1 Style Attributes

The game has a 16-bit aesthetic, using references from Super Nintendo Games such as The Legend of Zelda: A Link to the Past, The Escapists, Mario & Luigi Sagas or Pokemon. Each region of the map has a unique characteristic which gives a visual feedback to the player to know where is him. Style graphic is pixel art in 2D, with resolution of 16x16 bits for each tile.



*Illustrative Image from ToZ: A Link to the Past*



**Cards Display (bottom center):** The six cards available during the game will be displayed in the lower central area. These cards are pre-selected by the player. Cards can be selected using the mouse with a left-click. Selected cards will be highlighted with a blue border and the player can choose to use multiple cards at once

**Player health (bottom right) :** Displays the current health of the player

**Deck menu (bottom left):** This is the main menu of the game. When opened, it shows the player's deck, allowing them to select the six cards they will use in-game. The player can also enable or disable cards that function as passive abilities



*Illustrative Image: Player uses a card against an enemy*

Visual feedback will be used for all important actions occurring in the game. When the player takes damage, the screen will display a brief red flash. As the player's health decreases, the intensity of this flash will increase. Enemies, when taking damage, will briefly change their color to red for less than half a second

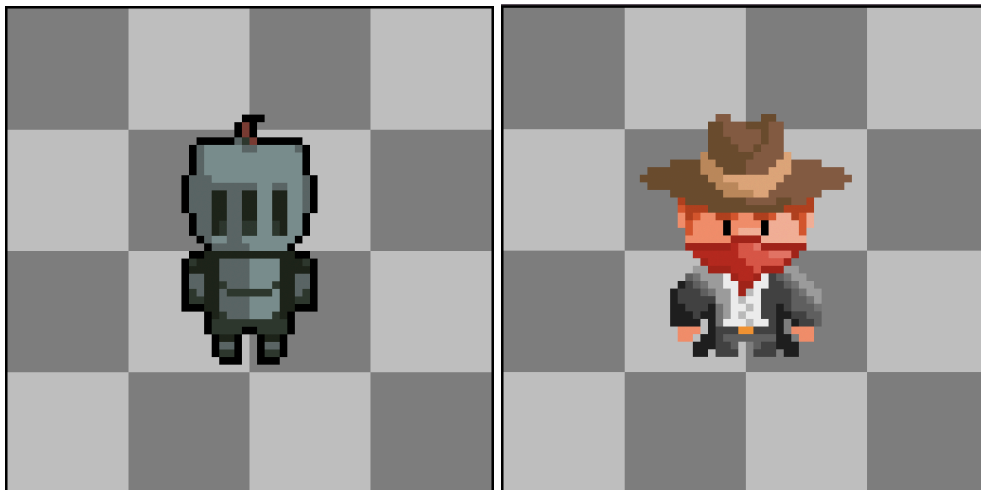
When a card is obtained, an image of the card will be displayed, drawing the player's attention as the main focal point



## 5.3 Sprites

### Player:

- Dimension: 32 x 32 px
- Idle in 4 directions: left, right, up, down
- Walking in 4 directions: left, right, up, down
- Take damage
- Death



### Enemies: Default movements

- Dimension: Depends on the enemy type
- Idle in 2 directions: left, right

- Walking in 2 directions: left, right
- Attacking in 2 directions: left, right



### Tiles:

Each dimension will have a base map with a resolution of 480 x 352 px. From this, a tileset will be used by the system to procedurally determine the placement and distribution of tile objects and decorations across the map

## 1. Old West

### a. Terrain

- Sand
- Dirt path
- Rocky ground
- Small elevation

### b. Natural props

- Cacti
- Dry bush
- Animal bones

### c. Interactive Objects

- Boxes

- ii. Chests

#### d. Obstacles

- i. Rocks

### 2. Dark Ages

#### a. Terrain

- i. Stone floor
- ii. Pillars
- iii. Doors

#### b. Natural props

- i. Torches
- ii. Tables
- iii. Blood
- iv. Bones

#### c. Interactive objects

- i. Chests





## 6. Sounds/Music

### 6.1 Style Attributes

The soundtrack is a core component of the game experience. It is built around a 16-bit aesthetic, using synthesized sound and short looping compositions that reinforce the repetitive style of a roguelite game

The primary goal of the soundtrack is to generate a constant sense of tension, communicating to the player that they are in a dangerous environment where enemies can appear at any moment and potentially end the game. The music does not evolve with the player progression, instead, it maintains sustained pressure. Each dimension has its own distinct soundtrack identifies to differentiate environments and reinforce immersion

**Old West:** Inspired by classic westerns and desert styles. The music conveys isolation, vast open spaces and latent danger. It uses slower tempos and conveys a sense of isolation, vastness and latent danger:

<https://www.youtube.com/watch?v=KhChHmk1o8c>

**Dark Age:** Inspired by dark fantasy dungeons settings and classic games such as The Legend of Zelda, conveying a sense of confinement and constant threat:

<https://www.youtube.com/watch?v=Nxx3Ti83TYk>

### 6.2 Sounds Needed

All sound effects follow a retro 16-bit aesthetic to maintain consistency with the visual and musical identity of the game. Sound effects are grouped based on their function:

**Interaction sounds:**

- Chest opening: <https://www.freesound.org/people/jon1/sounds/95503/>
- Impacts against walls or objects:  
<https://www.freesound.org/people/toxicwafflezz/sounds/150838/>

**Player and enemy feedback:**

- Damage effect (player): <https://www.freesound.org/people/mrickey13/sounds/515624/>
- Damage effect (enemy):  
<https://www.freesound.org/people/Mrthenoronha/sounds/506586/>

Each sound must clearly differentiate between player and enemy actions

**UI Sounds:**

- Card reveal
- Card played
- Menu selection <https://www.freesound.org/people/ZeltBolt/sounds/833055/>

**Ambient Sounds:**

- Desert wind <https://www.freesound.org/people/felix.blume/sounds/156414/>
- Whispering distant voices

Ambient sounds help reinforce the atmosphere of each dimension and should remain subtle to avoid overwhelming the player

**6.3 Music Needed**

The game requires different music tracks depending on the gameplay state

**Core Gameplay**

- Dungeon exploration <https://www.newgrounds.com/audio/listen/625484>
- Combat <https://www.newgrounds.com/audio/listen/76637>

## Key Event

- Final boss battle <https://www.newgrounds.com/audio/listen/436762>

## UI / Game States

- Main menu <https://www.playonloop.com/2020-music-loops/misty-dungeon/>
- Defeat screen <https://www.newgrounds.com/audio/listen/368887>
- Victory screen <https://www.newgrounds.com/audio/listen/1323003>
- Credits screen <https://www.newgrounds.com/audio/listen/11684>

## Special Content

- Special room

## 7. Schedule

The game development is organized into 8 stages and will be carried out in a staggered and progressive manner, maintaining a modular development that allows for system scalability with the aim of facilitating easier system debugging. Each system should only do one job and not several. The team consists of 3 developers, and collaborative work will be done with GitHub. The presented timeline maintains a clear and concise structure, prioritizing the individual development of each element for its integration at the end of each stage. The team seeks to perform continuous tests and adjustments for the early detection of problems that could hinder or compromise the complete development of the system.

### Week 1 – Core System

- Game object class development
- Implement of:
  - Player class (basic movements and initial attributes)
  - Enemies class

- Input system implementation
- Basic physics and collisions system
- Initial structure card system
- First version of the game loop

**Goal:** Establish foundation game and minimal playable game

## **Week 2 – Gameplay and card system**

- Completed card system development
  - Active card system
  - Passive card system
- Implementation of card mechanics (damage, effects)
- Enemy development (basic movement and logic)

**Goal:** Implement the main gameplay mechanics

## **Week 3 – Procedural Map Design**

- Development of the basic procedural map generator
- Implementation of:
  - Enemy AI
  - Store system (buying card)
  - Economic system (currency)
- Progression system and difficulty scaling
- Design of cards and enemies

**Goal:** Procedural map generator, game logic and dynamic content

## **Week 4 – UI / UX**

- UI development:
  - Game menu (main menu, pause menu)
  - Inventory system
  - Victory screen and Death screen
  - Deck UI (card management interface)
- UX feedback
  - Player visual indicator (damage, effects, interactions)
  - Player action feedback
- Integration of sound effects (combat, background sounds, actions)
- Refinement game loop
- Gameplay balancing
  - Gameplay flow
  - Player feelings
  - Cards balancing
  - Economy

**Goal:** Improve user experience and visual final finishes

### **Week 5 – Testing, debugging and final delivery**

- Gameplay testing
  - Internal testing
  - External player testing
- Bug fixing
- Development of website
- Final build and presentation